Huawei Access Node MALL-ACC-03 Routing Protocol Flap Resolution SOP

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_4338

Description

The Huawei access node MALL-ACC-03, operating within the Malleswaram Node site, experiences a minor routing protocol flap due to excessive BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This failure domain occurs on the specified role and vendor equipment due to the inherent complexities of routing protocol interactions, particularly when dealing with multi-protocol environments. The Huawei NE20E2 V5.21.01 router, utilized in this setup, employs a multi-process architecture, with each process responsible for managing a specific routing protocol instance. When BGP route churn occurs, it can lead to an increased load on the routing engine, causing OSPF adjacency loss and subsequent ISIS LSP flooding. This, in turn, exacerbates the BGP route churn, creating a vicious cycle. The root cause of this issue may lie in misconfigured routing policies, inadequate route filtering, or hardware limitations.

Pre-requisite Checks

- check: Verify the Malleswaram Node site is accessible and the NE20E2 V5.21.01 router is powered on.

- check: Ensure redundant links are active and functioning properly.

- check: Dispatch remote hands to Malleswaram Node for physical verification of equipment and cables.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policies	High
Inadequate route filtering on adjacent routers	Medium
Hardware limitations in the NE20E2 V5.21.01 router	Low

Diagnostic Commands & Expected Outputs

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Command: display bgp summary, Expected Output: BGP Summary: Router ID: 10.1.1.1 AS: 100 Total AS-Paths: 100 Total BGP Routes: 500 , Description: This command displays a summary of the BGP routing table, including the router ID, AS number, total AS-paths, and total BGP routes. It helps identify any anomalies in the BGP routing table.

Command: display ospf neighbor, Expected Output: OSPF Neighbor Table: Neighbor ID: 10.2.2.2 State: FULL Dead Time: 30 , Description: This command displays the OSPF neighbor table, including the neighbor ID, state, and dead time. It helps identify any OSPF adjacency loss
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issues.

Command: display isis lsp, Expected Output: ISIS LSP Table: LSP ID: 10.3.3.3 Sequence Number: 12345 , Description: This command displays the ISIS LSP table, including the LSP ID and sequence number. It helps identify any ISIS LSP flooding issues.
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Resolution Steps

Step 1: Step: Verify and correct BGP routing policies, Command: display bgp policy, Description: This command displays the BGP routing policies, which can be used to identify and correct any misconfigured policies causing the BGP route churn.

Step 2: Step: Verify and correct route filtering on adjacent routers, Command: display ip routing-table, Description: This command displays the IP routing table, which can be used to identify and correct any inadequate route filtering on adjacent routers.

Step 3: Step: Upgrade or replace the NE20E2 V5.21.01 router, Command: system-view upgrade system, Description: This command upgrades or replaces the NE20E2 V5.21.01 router, which can resolve hardware limitations causing the issue.

Post-Check Verification

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Command: display bgp summary, Expected Output: BGP Summary: Router ID: 10.1.1.1 AS: 100 Total AS-Paths: 100 Total BGP Routes: 500 , Description: This command verifies that the BGP routing table has stabilized and there are no anomalies.

Command: display ospf neighbor, Expected Output: OSPF Neighbor Table: Neighbor ID: 10.2.2.2 State: FULL Dead Time: 30 , Description: This command verifies that the OSPF adjacency has been restored and there are no issues.

Command: display isis lsp, Expected Output: ISIS LSP Table: LSP ID: 10.3.3.3 Sequence Number: 12345 , Description: This command verifies that the ISIS LSP table has stabilized and there are no issues.
```

Roll-back Procedures

- Step: Restore the previous BGP routing policies, Command: system-view undo bgp policy, Description: This command restores the previous BGP routing policies, which can be used to roll back the changes made during the resolution process.

- Step: Restore the previous route filtering on adjacent routers, Command: undo ip routing-table, Description: This command restores the previous route filtering on adjacent routers, which can be used to roll back the changes made during the resolution process.